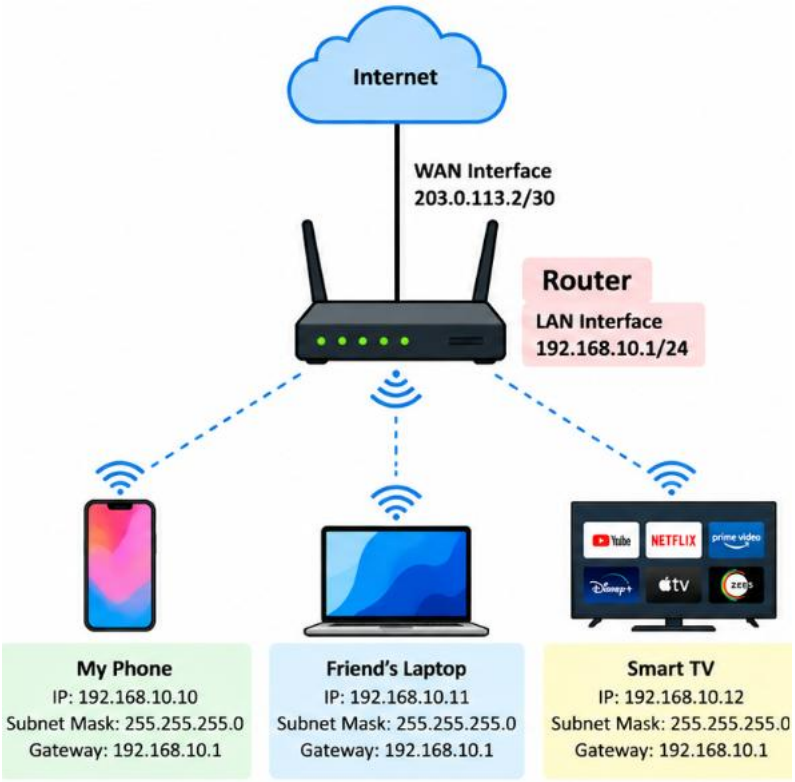
IP Routing and Routing Protocols

1. Network Diagram with IP Addresses

The following diagram shows three devices connected through a router. All devices are in the same local network, 192.168.10.0/24.

Subnet mask for all three devices: 255.255.255.0

Default gateway for all three devices: 192.168.10.1

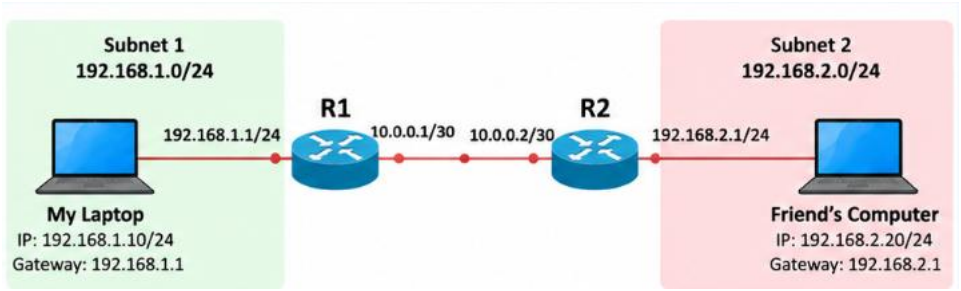


The router's LAN interface connects the devices to the local network. Its WAN interface connects to the ISP or an upstream network. The WAN address shown is an example documentation address, not a real public service address.

2. How a Packet Travels Between Different Subnets

Suppose my laptop needs to send data to my friend's computer, which is on a different subnet.

Network diagram



Step-by-step process

1. Check the destination: My laptop compares the destination IP 192.168.2.20 with its own subnet, 192.168.1.0/24. It determines that the destination is on a different subnet.
2. Send to the gateway: My laptop sends the packet to its default gateway, 192.168.1.1.
3. Check the routing table: Router R1 checks its routing table and finds that 192.168.2.0/24 is directly connected to its second interface.
4. Forward the packet: R1 sends the packet out through interface 192.168.2.1, using the appropriate link-layer address for my friend's computer.
5. Receive the packet: My friend's computer receives the packet addressed to 192.168.2.20.

Routing table checked by R1

Destination Network	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Directly connected	G0/0
192.168.2.0	255.255.255.0	Directly connected	G0/1

How the decision is made: The router selects the most specific matching route. Here, both networks are directly connected, so R1 forwards the packet through the interface connected to the destination subnet.

3. Routing Table for a Zomato-Style Delivery Network

Imagine a food delivery company with three branches in Ahmedabad, Mumbai and Pune. Each branch has its own IP subnet, and the headquarters router forwards traffic to the appropriate branch router.

Assume the headquarters router is named HQ-R1.

Branch	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad	10.1.1.0/24	255.255.255.0	10.0.0.2	Serial0/0/0
Mumbai	10.2.2.0/24	255.255.255.0	10.0.0.6	Serial0/0/1
Pune	10.3.3.0/24	255.255.255.0	10.0.0.10	Serial0/1/0

These next-hop addresses represent the neighboring routers on separate point-to-point links. The interface names are illustrative and would need to match the actual router configuration.

Explanation: When HQ-R1 receives a packet, it checks the destination IP address and selects the matching route. It then forwards the packet to the appropriate next-hop router through the specified interface.

4. Static Routing vs. Dynamic Routing

Type	Advantages (3)	Disadvantages (3)
Static Routing	1. Simple for small networks. 2. No routing-protocol update traffic. 3. Gives the administrator direct control over routes.	1. Routes must be updated manually. 2. Does not automatically adapt to a failed link. 3. Becomes difficult to manage as the network grows.
Dynamic Routing (RIP/OSPF)	1. Learns routes automatically. 2. Can find alternate paths when network conditions change. 3. Suitable for larger networks with multiple branches.	1. Requires more configuration and monitoring. 2. Uses CPU, memory and network resources. 3. Convergence after a failure takes time and depends on the protocol.

Real-world example

- Static routing: A small restaurant with one router and one Internet connection may use a default static route.
- Dynamic routing: A large food delivery company with multiple branches and redundant links could use OSPF to learn routes and adapt when a link fails.

Conclusion: Static routing works well for simple, stable networks, while dynamic routing is more suitable for growing networks that need automatic route updates and alternate paths.

5. Sample OSPF Configuration Between Two Routers

OSPF (Open Shortest Path First) is a dynamic routing protocol that allows routers to exchange routing information and calculate suitable paths automatically.

Network diagram

Router R1

192.168.1.1/24

Link network: 10.0.0.0/30

Router R2

192.168.2.1/24

The routers use the 10.0.0.0/30 network for their connection:

- R1 link interface: 10.0.0.1/30
- R2 link interface: 10.0.0.2/30

Configuration for Router R1

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

Configuration for Router R2

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

These commands assume the stated interface names exist on the routers and the two routers are connected through those interfaces.

Explanation of the commands

Command	Purpose
enable	Enters privileged EXEC mode.
configure terminal	Opens global configuration mode.
interface gigabitEthernet 0/0	Selects the specified interface.
ip address ...	Assigns an IP address and subnet mask.
no shutdown	Activates the interface.
exit	Returns to the previous configuration mode.
router ospf 1	Starts OSPF process 1.
network ... area 0	Enables OSPF on matching interfaces and advertises their networks in area 0.
end	Returns to privileged EXEC mode.

The wildcard mask 0.0.0.255 matches a /24 network, while 0.0.0.3 matches a /30 network.

How to verify OSPF

After configuring both routers, run:

show ip ospf neighbor

show ip route ospf

show ip protocols

- show ip ospf neighbor checks whether the routers have formed an OSPF neighbor relationship.
- show ip route ospf displays routes learned through OSPF.
- show ip protocols displays routing protocol information.

Expected result: R1 should learn the 192.168.2.0/24 network through R2, and R2 should learn 192.168.1.0/24 through R1, provided the interfaces are connected and OSPF adjacency forms successfully.